

Regression Analysis of Medical Insurance Claim Amounts

Salma Abdallah

Course: STAT 311 Regression Analysis, Spring 2026

Date: April 28, 2026

This project analyzes the relationship between medical insurance claim amounts and several demographic and health-related variables. The main research question is: How do factors such as age, BMI, smoking status, diabetes status, and gender influence medical insurance claim amounts, and which variables are the strongest predictors of healthcare costs?

Understanding these relationships is important because healthcare costs are a major financial concern for individuals and insurance providers. Identifying key drivers of insurance claims can help improve pricing, risk assessment, and policy decisions.

To answer this question, multiple regression analysis was performed using JMP. The analysis included exploratory data analysis, model building, testing higher order terms, stepwise regression for model selection, and evaluation of model assumptions. A log transformation of the response variable was also considered to improve model performance.

The dataset used in this project is the Real U.S. Medical Insurance Costs Dataset, obtained from Kaggle. It contains 4,000 observations and includes variables describing individuals demographic and health characteristics along with their insurance claim amounts.

The variables used in this analysis include:

- Age (continuous)
- Gender (categorical)
- BMI (continuous)
- Smoking status (categorical)
- Diabetes status (categorical)
- Claim_Amount (continuous, response variable)

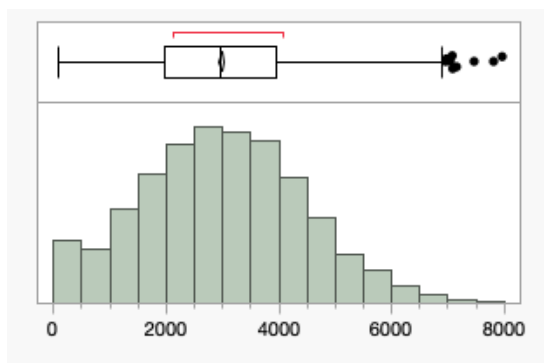
- Claim status (categorical)

A log transformed version of the response variable (Log_Claim) was also created to address skewness.

The data generating process reflects real world variation in healthcare costs based on individual characteristics. It is expected that factors such as health conditions and lifestyle behaviors may influence claim amounts. However, the dataset does not include all possible drivers of healthcare costs, which may limit the model's explanatory power.

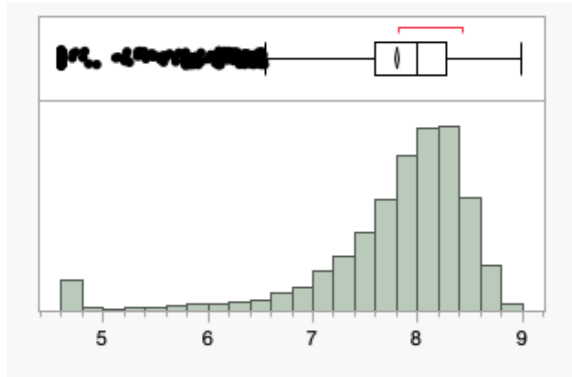
Exploratory analysis was conducted to understand the distribution of the variables and identify potential relationships.

Figure 1: Distribution of Claim Amount



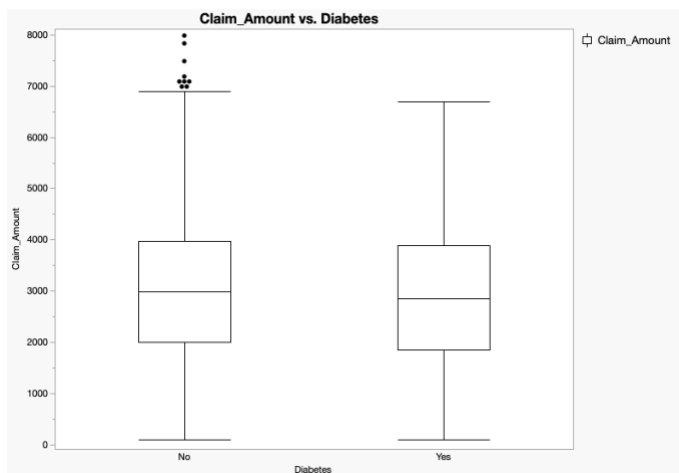
The distribution of claim amounts is right skewed, with most values concentrated between approximately \$1,500 and \$4,500, and a few higher values extending beyond \$6,000. This indicates the presence of high cost outliers.

Figure 2: Distribution of Log Claim Amount



After applying a log transformation, the distribution becomes more symmetric. This reduces skewness and improves suitability for regression modeling.

Figure 3: Claim Amount vs Diabetes



The boxplot shows differences in claim amounts between individuals with and without diabetes. While there is overlap, the distributions suggest some variation between the groups.

Figure 4: Claim Amount vs Age

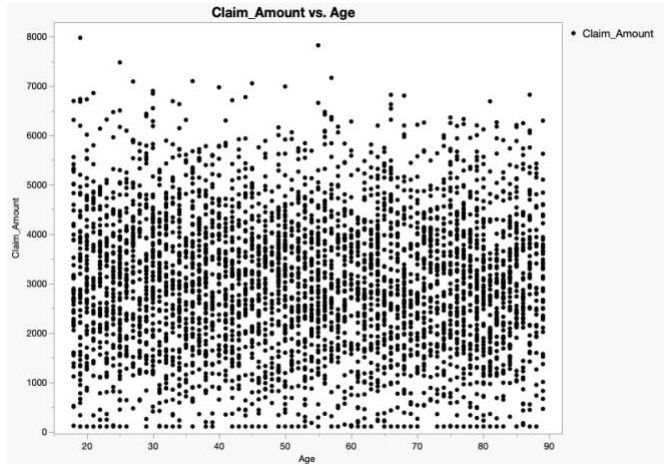


Figure 4 shows the relationship between age and claim amount. The scatterplot displays a wide spread of data points across all age values, with no clear linear trend. Claim amounts vary significantly at every age level, indicating that age does not have a strong influence on insurance claim costs.

This observation is consistent with the regression results, where age was not found to be a statistically significant predictor.

Figure 5: Claim Amount vs BMI

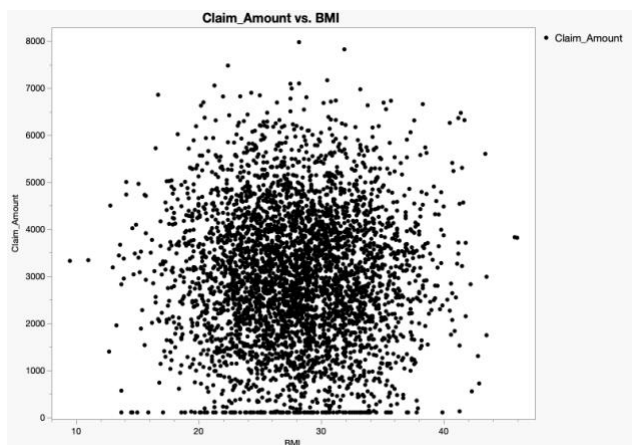
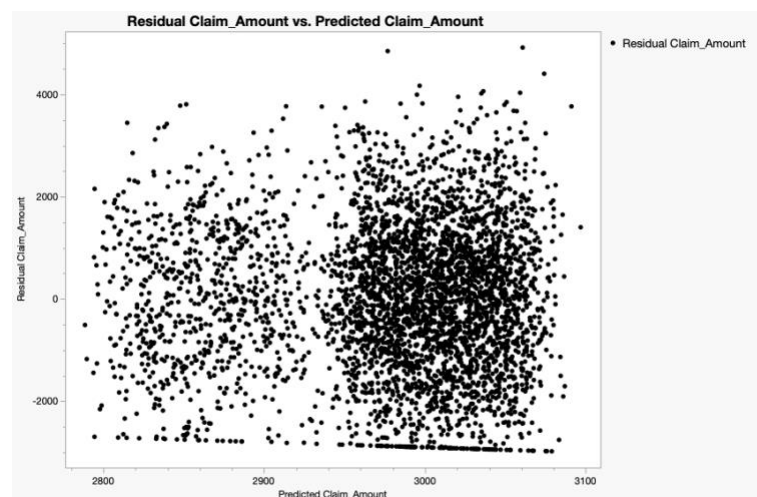


Figure 5 shows the relationship between BMI and claim amount. The scatterplot displays a wide dispersion of data points with no clear linear pattern. Claim amounts vary significantly across all BMI values, indicating that BMI does not have a strong relationship with insurance claim costs.

This finding is consistent with the regression analysis, where BMI was not found to be a statistically significant predictor.

Figure 6: Residual Plot (Residuals vs Predicted Claim Amount)



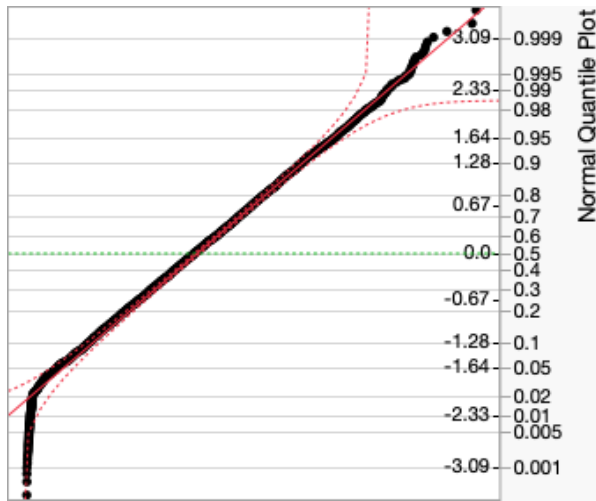
A residual plot of residuals versus predicted values was generated to assess model assumptions.

The plot shows that residuals are roughly centered around zero with no clear systematic or nonlinear pattern, suggesting that the linearity assumption is reasonably satisfied for this model.

However, the presence of vertical clustering indicates that the predicted values take on only a limited number of values, which reflects the limited complexity of the model and the small number of predictors retained after model selection. This occurs because the final model includes only a single categorical predictor (diabetes).

Additionally, the wide spread of residuals suggests high variability and indicates that the model does not explain a substantial portion of the variation in claim amounts, which is consistent with the very low R^2 value observed in the regression results.

Figure 7: Normal Q-Q Plot of Residuals



A normal quantile (Q-Q) plot of the residuals was created to assess the normality assumption. The majority of the points lie close to the reference line, indicating that the residuals are approximately normally distributed. Some deviation is observed in the lower tail, suggesting the presence of a few extreme values; however, overall the normality assumption is reasonably satisfied.

This study uses an observational dataset, meaning that no experimental manipulation or controlled treatment was applied to the data. Instead, the analysis relies on naturally occurring variation in the variables to investigate relationships between predictors and the response variable. The primary goal of this study is to understand how different demographic and health related factors influence medical insurance claim amounts. To achieve this, multiple linear

regression was used to model the relationship between claim amount and a set of selected predictors. The initial regression model included the following predictors: Age, Body Mass Index (BMI), Gender, Smoking status, and Diabetes status. These variables were chosen based on their relevance to healthcare costs and their potential influence on insurance claims. To satisfy the project requirement of including at least six predictors, Claim Status was also incorporated into the model as an additional categorical variable, even though it may not be a primary driver of claim amounts. To capture more complex relationships in the data, additional modeling techniques were applied. Higher order terms for Age and BMI (Age^2 and BMI^2) were included to account for possible nonlinear effects, such as situations where claim amounts increase at different rates across age groups or BMI levels. Furthermore, an interaction term between BMI and Smoking status was added to examine whether the effect of BMI on claim amount differs depending on whether an individual is a smoker or non-smoker.

Categorical variables, including Gender, Smoking status, Diabetes status, and Claim Status, were treated as nominal variables. JMP automatically encoded these variables using indicator variables, allowing them to be properly included in the regression model. This encoding ensures that differences between categories are captured in the model estimates. Model selection was performed using stepwise regression with backward elimination based on p-value thresholds. This approach starts with a full model that includes all selected predictors and systematically removes variables that are not statistically significant. The goal of this process is to identify a more parsimonious model that includes only the most relevant predictors while avoiding unnecessary complexity. Model adequacy was evaluated using several statistical measures. R^2 and Adjusted R^2 were used to assess how well the model explains the variability in claim amounts. P-values were examined to determine the statistical significance of each predictor in

the model. In addition, residual analysis was conducted to check key regression assumptions, including linearity, constant variance, and normality of residuals. To further address potential violations of model assumptions, a log transformation of the response variable was applied. This transformation helps reduce skewness in the distribution of claim amounts and improves the normality of residuals, which is important for the validity of regression inference.

Finally, model validation was conducted using a train/test split approach. The dataset was divided into two subsets: one used to train the model and another used to test its predictive performance. This approach provides a more realistic evaluation of how well the model generalizes to new, unseen data and helps ensure that the results are not overly dependent on the specific sample used for model fitting.

The relationship between medical insurance claim amounts and several demographic and health-related variables was analyzed using multiple linear regression. The response variable was Claim_Amount, and the predictors included Age, Body Mass Index (BMI), Gender, Smoking status, and Diabetes status. An additional categorical variable, Claim Status, was included in the model to satisfy the requirement of having at least six predictors.

The initial regression model was not statistically significant (overall F-test p-value > 0.05), indicating that the predictors, when considered together, do not provide strong evidence for explaining variation in claim amounts. The coefficient of determination (R^2) was approximately 0.002, meaning that less than 1% of the variability in claim amounts is explained by the model. This indicates extremely weak explanatory power.

At the individual predictor level, most variables were not statistically significant (p -values > 0.05), including Age, BMI, Gender, and Smoking status. Diabetes status was the only variable that consistently showed statistical significance, suggesting a measurable though limited relationship with claim amounts. To investigate potential nonlinear relationships, higher order terms (Age^2 and BMI^2) and an interaction term ($\text{BMI} \times \text{Smoking}$) were incorporated into the model. These additions were intended to capture curvature and interaction effects between predictors. However, the inclusion of these terms did not significantly improve model performance. The additional predictors were not statistically significant, and the R^2 value remained very low. This suggests that nonlinear effects and interactions do not meaningfully explain variation in claim amounts within this dataset.

Stepwise regression using backward elimination was then applied to identify a tighter model. The results confirmed that most predictors do not contribute meaningfully to explaining claim amounts. Across multiple iterations, Diabetes status was the only variable consistently retained as statistically significant. This reinforces the conclusion that, among the variables considered, diabetes is the only predictor with a detectable relationship to claim amounts.

Due to the right skewed distribution of claim amounts observed during exploratory data analysis, a log transformation of the response variable (Log_Claim) was applied to improve model assumptions, particularly the normality of residuals. While this transformation reduced skewness and produced a more symmetric distribution, it did not substantially improve the regression results. The R^2 value remained very low (approximately 0.001), and most predictors remained statistically insignificant. Diabetes status continued to be the only significant predictor ($p < 0.05$).

Based on the analysis, the final regression model can be expressed as:

$$\text{Claim_Amount} = 2936.44 - 71.92(\text{Diabetes})$$

This model suggests that individuals with diabetes have, on average, slightly lower claim amounts compared to those without diabetes. However, the magnitude of this effect is small, and the overall model explains very little of the variation in claim amounts, limiting its practical usefulness.

This study examined how demographic and health related variables influence medical insurance claim amounts using multiple linear regression analysis. The results showed that most of the variables considered, including age, BMI, gender, and smoking status, were not statistically significant predictors of claim amounts. Across all models tested, diabetes status was the only variable that was consistently identified as statistically significant.

In simple terms, these findings suggest that the dataset does not provide strong evidence for explaining why insurance claim amounts vary between individuals. While diabetes shows some relationship with claim amounts, the overall model explains very little of the variation, as indicated by very low R^2 values. This means that the variables included in the analysis are not sufficient to capture the key factors that drive healthcare costs. Several modeling approaches were explored in an attempt to improve the model. These included adding higher order terms, incorporating interaction effects, applying stepwise regression for model selection, and performing a log transformation of the response variable to address skewness. Although the log transformation improved the distribution of the data, none of these approaches led to a

meaningful improvement in model performance. The results remained consistent, with most predictors showing weak or no relationship with claim amounts.

One important implication of these findings is that the limitation lies primarily in the dataset rather than the statistical methods used. The dataset does not include several important variables that are likely to influence healthcare costs, such as detailed medical history, severity of conditions, income level, type of insurance coverage, or access to healthcare services. Without these variables, the model is unable to fully explain the variation in claim amounts.

From a practical perspective, this means that predicting insurance claim amounts based solely on basic demographic and health indicators may not be sufficient. More comprehensive data would be required to build a reliable predictive model.

In future work, the analysis could be improved by incorporating additional variables that better capture individual health risks and financial factors. Exploring alternative datasets or combining multiple data sources may also provide more meaningful insights. Additionally, more advanced modeling techniques could be considered if richer data becomes available.

Overall, this project demonstrates that even when appropriate statistical methods are applied correctly, the quality, relevance, and completeness of the data play a critical role in determining the success of a model. The results highlight the importance of carefully selecting datasets and recognizing the limitations of available data when conducting statistical analysis.

Appendix

1. Regression Output

Table A1: Summary of Fit

Summary of Fit	
RSquare	0.002201
RSquare Adj	0.000702
Root Mean Square Error	1428.36
Mean of Response	2979.707
Observations (or Sum Wgts)	4000

Table A2: Parameter Estimates

Parameter Estimates				
Term	Estimate	Std Error	t Ratio	Prob> t
Intercept	3067.5609	144.4281	21.24	<.0001*
Age	-1.572031	1.095911	-1.43	0.1515
BMI	-1.873409	4.509966	-0.42	0.6779
Gender[Female]	-1.130849	22.6113	-0.05	0.9601
Smoking[No]	-4.842137	24.65043	-0.20	0.8443
Diabetes[No]	71.659456	28.2789	2.53	0.0113*
Claim_Status[Approved]	10.189518	31.76394	0.32	0.7484

Table A3: Analysis of Variance (ANOVA)

Analysis of Variance				
Source	DF	Sum of Squares	Mean Square	F Ratio
Model	6	17973604.6	2995601	1.4683
Error	3993	8146573181	2040214	Prob > F
C. Total	3999	8164546786		0.1849

2. Stepwise Regression

Table B1: Stepwise Selection Results (Step History)

Step History									
Step	Parameter	Action	"Sig Prob"	Seq SS	RSquare	Cp	p	AICc	BIC
1	All	Entered	.	.	0.0022	7	7	69474.8	69525.1
2	Gender{Female-Male}	Removed	0.9601	5103.097	0.0022	5.0025	6	69472.8	69516.8
3	Smoking{No-Yes}	Removed	0.8442	78806.76	0.0022	3.0411	5	69470.8	69508.6
4	Claim_Status{Denied-Approved}	Removed	0.7444	216803.7	0.0022	1.1474	4	69468.9	69500.4
5	BMI	Removed	0.6811	344307.3	0.0021	-0.684	3	69467.1	69492.3
6	Age	Removed	0.1550	4123081	0.0016	-0.663	2	69467.1	69486

Figure C1: Residuals vs Predicted Values

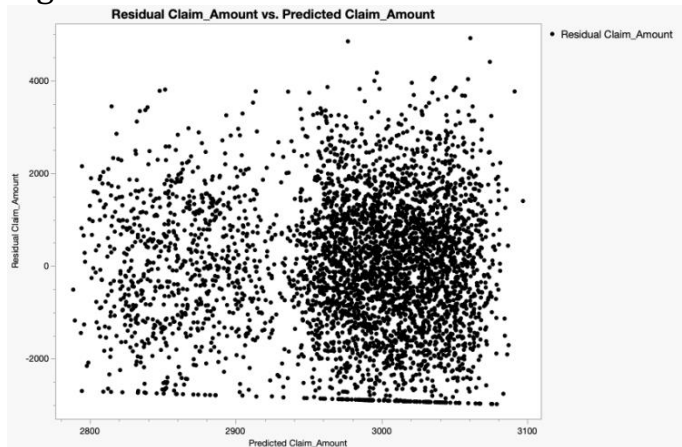


Figure C2: Normal Q-Q Plot of Residuals

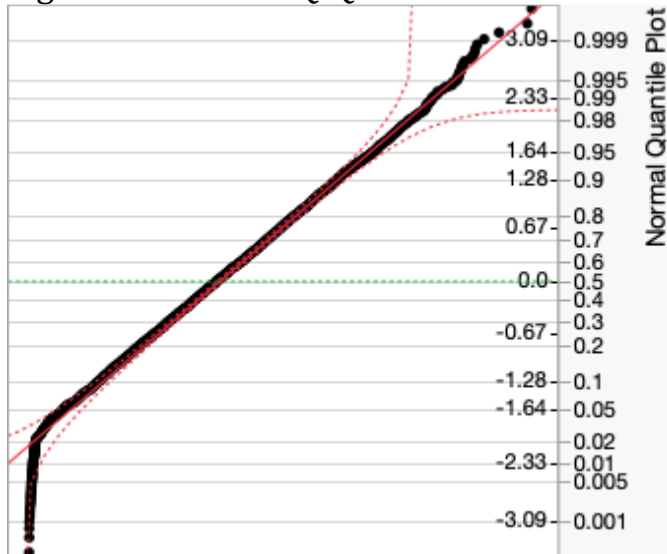


Table D1: Stepwise Regression Step History

Step History									
Step	Parameter	Action	"Sig Prob"	Seq SS	RSquare	Cp	p	AICc	BIC
1	All	Entered	.	.	0.0022	7	7	69474.8	69525.1
2	Gender{Female-Male}	Removed	0.9601	5103.097	0.0022	5.0025	6	69472.8	69516.8
3	Smoking{No-Yes}	Removed	0.8442	78806.76	0.0022	3.0411	5	69470.8	69508.6
4	Claim_Status{Denied-Approved}	Removed	0.7444	216803.7	0.0022	1.1474	4	69468.9	69500.4
5	BMI	Removed	0.6811	344307.3	0.0021	-0.684	3	69467.1	69492.3
6	Age	Removed	0.1550	4123081	0.0016	-0.663	2	69467.1	69486
7	Diabetes{Yes-No}	Removed	0.0110	13205502	-0.000	3.8097	1	69471.6	69484.2
8	Diabetes{Yes-No}	Entered	0.0110	13205502	0.0016	-0.663	2	69467.1	69486

Table D2: Final Model Estimates

Current Estimates							
Lock	Entered	Parameter	Estimate	nDF	SS	"F Ratio"	"Prob>F"
☒	☒	Intercept	2936.44471	1	0	0.000	1
☐	☐	Age	0	1	4123081	2.023	0.15503
☐	☐	BMI	0	1	310154.6	0.152	0.69657
☐	☐	Gender{Female-Male}	0	1	13151.7	0.006	0.936
☐	☐	Smoking{No-Yes}	0	1	93711.86	0.046	0.83027
☐	☒	Diabetes{Yes-No}	-71.923337	1	13205502	6.477	0.01097
☐	☐	Claim_Status{Denied-Approved}	0	1	151577.8	0.074	0.78515

References

Kaggle Dataset:

Real U.S. Medical Insurance Costs Dataset

<https://www.kaggle.com/datasets/mohsin07/real-u-s-medical-insurance-costs-dataset-csv>